

vapour deposition (MOCVD) tools to deposit graphene directly onto semiconductor wafers, bypassing the need for pre-made sheets. Metrology techniques like Raman spectroscopy and atomic force microscopy (AFM) ensure purity and performance directly on the wafer. The wafers then undergo semiconductor processes like dicing, wire bonding, and packaging, with every graphene-based device tested to meet the demands of applications ranging from vehicle sensors to smartphone chips.”

Role of graphene in battery management systems

Graphene, recognised for its exceptional electrical conductivity and versatility, is poised to advance the electric vehicle (EV) industry significantly, particularly in enhancing battery management systems (BMS). This innovative material is employed in sensors that enable improved current measurement, allowing for a more accurate assessment of the battery’s state of charge. This precision optimises charging while extending the battery’s range and lifespan. “There is ongoing, extensive collaboration between manufacturers and leading automotive suppliers to integrate graphene-based sensors into BMS effectively,” adds John.

Considering the costs, you might wonder if such high-tech solutions are financially feasible. Interestingly, incorporating graphene into BMS is cost-neutral at the system level. The approach here involves replacing expensive shunt resistors with graphene sensors. This swap balances out the financial equation and enhances performance without adding extra costs. System designers are particularly enthusiastic about this update, seeing it as a win-win for enhancing system capabilities without straining budgets.

One transformative aspect of using graphene in BMS is eliminating

Superior performance in cryogenic conditions

Graphene excels in cryogenic temperatures where traditional materials like silicon fail:

- **Robust at ultra-low temperatures.** Unlike semiconductors that stop working near absolute zero, graphene remains functional, serving as a Hall effect sensor at low temperatures.
- **Broad temperature responsiveness.** Graphene ensures consistent sensor performance across a wide temperature range, which is ideal for magnetic sensing and other applications.
- **Simplified temperature calibration.** Graphene enables easy and uniform temperature calibration with its atomic structure and lack of a bandgap.
- **Versatility in sensing applications.** Its structure and broad temperature range make graphene suitable for various detection technologies where conventional materials fall short. Graphene’s properties make it a reliable choice for extreme environments and advanced technological applications.

traditional shunt resistors notorious for generating heat and handling high currents—typically between 600 and 2000 amps. These resistors require very low resistance, which can compromise accuracy at lower currents. Replacing these with graphene-based magnetic sensors reduces heat generation and improves accuracy at those crucial lower currents.

This shift represents a leap to more robust, reliable, and efficient BMS designs. The industry sets new standards by turning to graphene, demonstrating how innovative materials can lead to more sustainable and effective technologies in the automotive sector.

Is graphene the future of data centres?

In the dynamic world of data centres, the perpetual challenge is to boost computing power while simultaneously cutting energy costs. Graphene and other 2D materials are becoming pivotal, transforming multiple levels of system architecture. At the heart of the hardware, these materials are redefining the efficiency of transistors. Their application in memory systems is particularly promising, especially for AI applications, which are notoriously memory-intensive. The potential for these materials to drastically reduce energy consumption in memory architecture is a thrilling development.

As we scale up from individual components to entire systems, graphene’s impact becomes even more apparent. In communications technology within data centres, graphene can enhance the performance of modulators and receivers. Moreover, it can improve the efficiency of power supplies through the integration of graphene-based sensors. These widespread improvements underscore the extensive potential of graphene in enhancing data centre operations.

The immediate challenge lies in proving that graphene can effectively create higher-performing, energy-efficient transistors and memory systems. The memory sector, in particular, might present a more straightforward avenue for integrating these novel materials compared to the complex task of overhauling existing processing methods. Such integration could mark a significant advancement in evolving data centre technologies.

John says, “Our focus remains on exploiting graphene in applications where it retains its one-atom-thick form. While graphene shows promise in more voluminous three-dimensional structures, such as inks and printable media known for their excellent thermal properties and conductivity, these are not the main focus of our research. Instead, we concentrate on leveraging graphene in semiconductor devices, where